

Adaptive-Persistence Online Platt Scaling

A clean algorithmic frame and the minimum experiment needed to decide whether it works

Working name: AP-OPS | September 2026

CENTRAL PROPOSAL

Keep full Online Platt Scaling as the calibration primitive, but make its temporal memory adaptive. AP-OPS runs one reset expert and two persistent experts in parallel, then combines their probabilities using only matured feedback. The reset expert is the exact current OPS implementation, so the new method remains anchored to the strongest observed calibrator.

1. Why this is the right next method

The completed experiments give a sharp design signal. Full online Platt scaling is the strongest method on Criteo (log loss 0.606958 versus 0.607070 for DualTime-CTR) and is close to the best result on Avazu (0.387443 versus 0.387402 for AdaMoE). The learned slope matters: fixing a at one removes an important degree of correction. At the same time, Avazu diagnostics suggest that information from the previous day may help at the start of a new day, while Criteo favors resetting.

This points to one focused extension: preserve the full two-parameter map, expose only a few distinct memory scales, and learn online which memory scale to trust. The method adds temporal adaptation without returning to a high-dimensional contextual residual model.

Proposed contribution

Layer	What changes
Calibration	Every expert uses the full monotone Platt map, so both slope and intercept adapt.
Temporal memory	Experts span reset, short persistence, and long persistence.
Online selection	A delayed fixed-share rule shifts weight among memory scales as the regime changes.
Guarantee	The aggregate competes with the current OPS anchor and with changing sequences of memory experts.

PAPER-LEVEL FRAME

The novelty is the **adaptive temporal-memory strategy**. Standard OPS is introduced and cited once as the underlying online calibration primitive; AP-OPS is then presented as the new extension.

2. The AP-OPS algorithm

Let q_i be the frozen cross-day CTR probability for impression i . All experts receive the same ordered stream and the same delayed labels. Define the clipped base logit and expert prediction by:

```
z_i = logit(clip(q_i, eps, 1 - eps)), eps = 1e-5
p_i^(k) = sigmoid(a_r^(k) z_i + b_r^(k)), a in [0.2, 5], b in [-0.25, 0.25]
p_i^AP = sum_k w_r,k p_i^(k).
```

A deliberately small expert bank

Expert	State across days	Role
R: reset anchor	Reset (a,b) to (1,0) each day; use the current projected-gradient update.	Protects the verified empirical OPS behavior.
S: short memory	Carry state continuously; discounted ONS with a 4-hour half-life.	Tracks rapid within-day or day-boundary shifts.
L: long memory	Carry state continuously; discounted ONS with a 16-hour half-life.	Retains stable calibration information across days.

Delayed second-order expert update

For block r , compute its mean log loss only after the configured feedback delay. The two persistent experts update the 2×2 curvature matrix and the Platt parameters by discounted Online Newton Step:

```
g_r = grad_theta L_r(theta_r)
A_r = gamma A_(r-1) + g_r g_r^T + (1 - gamma) lambda I
theta_(r+1) = Proj_Theta[A_r](theta_r - eta_ons A_r^(-1) g_r),
where gamma = 2^(-Delta_t / h).
```

Delayed adaptive aggregation

Initialize the expert weights at **(0.50, 0.25, 0.25)** for (R,S,L). The weights persist across days. When a block becomes observable, update them using that block's pre-update expert losses and then apply fixed share:

```
w_tilde_(r+1,k) proportional to w_(r,k) exp(-eta_m L_(r,k))
w_(r+1,k) = (1 - alpha) w_tilde_(r+1,k) + alpha / 3.
```

AP-OPS: one chronological block	Causality rule
1. Emit R, S, and L predictions; output their probability mixture.	Use current states and weights only.
2. Queue the block without changing any emitted prediction.	No unmaturing label enters the state.
3. When feedback matures, score each expert on its stored prediction.	Loss precedes every update.
4. Update expert states, then update and share the meta-weights.	The new state applies only to later blocks.
5. At a day boundary, reset R; carry S, L, and the meta-weights.	This is the only expert-specific reset.

3. Minimal experiment needed to decide whether AP-OPS works

Do not launch a broad algorithm sweep. Reuse the existing cross-day predictions and replay only the calibration layer. The new study needs one main comparison and two component ablations.

Row	What is run	Question answered
Current OPS	Exact locked implementation: full (a,b), projected gradient, daily reset.	Can the new method beat or safely match the current empirical anchor?
AP-OPS	The three-expert algorithm above, with all choices frozen on dev.	Does adaptive persistence work?
Single-memory ablation	Use the better of S and L selected on dev, without the meta-mixture.	Is adaptation across memory scales useful, rather than one persistent calibrator?
No-slope ablation	Run AP-OPS with a fixed at 1 and update only b.	Is the slope degree of freedom essential to the new method?

WHAT IS REUSED

The existing Expanding, Best Fixed Window, ARW, AdaMoE, and DualTime-CTR results remain in the paper's comparison table. They do not need another tuning sweep. Only current OPS is rerun as a golden reference on the exact stream used by AP-OPS.

Evaluation protocol

Phase	Required action
Development	Use the established dev days and seeds 0, 1, 2. Tune only ONS scale/regularization and the two meta parameters (learning rate and share rate). Freeze one configuration before evaluation.
Golden check	Require the reset expert, when run alone, to reproduce the saved OPS result within numerical tolerance. Require an anchor-only AP-OPS wrapper to be prediction-equivalent.
Fixed test	Evaluate the four rows once on the existing locked stream for direct comparability.
Rolling origin	Evaluate the frozen rows at every established chronological origin. This is the primary evidence for temporal adaptation and day-level inference.

Metrics that matter

- **Primary:** impression-weighted log loss and paired calendar-day difference relative to current OPS.
- **Temporal:** loss before the first feedback arrives, first-quarter-of-day loss, and worst-day loss.
- **Mechanism:** expert-weight paths and the number of days or blocks on which each memory scale dominates.
- **Secondary:** Brier score, runtime, and memory overhead. These do not determine method selection.

For inference, average the three seeds within each calendar day first. Bootstrap or randomize over days, not over impressions or seed-day pairs. Show every paired daily difference when the number of days is small.

Leakage and stability checks

- Perturbing future or unmaturred labels must leave all earlier predictions, parameters, and weights unchanged.
- Weights must remain nonnegative and sum to one; parameters must remain inside the stated bounds.
- The same seed and cached base-prediction stream must reproduce identical outputs.

4. Decision rule, theory target, and paper claim

A predeclared decision rule

Set the non-inferiority margin on development data before the final run:

$$\text{delta_NI} = 0.10 \times | \text{loss_OPS,dev} - \text{loss_base-q,dev} |.$$

Outcome	Evidence required	Interpretation
Works	AP-OPS is non-inferior to OPS on both datasets, improves the point estimate on at least one, and has no material early-day or worst-day regression. A rolling-day interval below zero supports a strict improvement claim.	Advance AP-OPS; calibrate the claim to the interval.
Retains performance	AP-OPS is non-inferior but not significantly better; expert weights adapt as intended and the comparator theorem holds.	Lead with adaptive memory and the stronger guarantee.
Failure	AP-OPS exceeds the non-inferiority margin or shows a systematic early-day or worst-day regression.	Keep current OPS; report AP-OPS only as an ablation.

Theory target for the exact implemented algorithm

At the base level, binary log loss over the compact two-parameter Platt family is exp-concave after clipping. Each ONS expert should therefore admit logarithmic static regret in dimension two. Positivity of α preserves the ranking induced by the base CTR score.

At the meta level, analyze the actual delayed fixed-share update on mean block losses. The target statement is a switching-comparator bound of the form:

$$\text{Loss_AP}(T) \leq \min \text{ over expert paths } k_{1:T} \text{ with } S \text{ switches} \\ \sum_r \text{Loss_}(r, k_r) + \text{Meta}(T, K=3, S) + \text{Delay}(T, D).$$

Because the current OPS implementation is expert R, the case $S = 0$ immediately yields a comparator guarantee against the empirical anchor, up to the explicit meta and feedback-delay terms. The changing-expert case formalizes the intended advantage under temporal shifts. The proof must use the same block averaging, clipping, update order, and delay convention as the code.

Recommended contribution statement

DRAFT LANGUAGE

We introduce Adaptive-Persistence Online Platt Scaling, a delayed-feedback calibration strategy that preserves full slope-and-intercept correction while adapting its temporal memory across reset, short-memory, and long-memory experts. The method competes with the verified OPS anchor and with changing memory scales, and is evaluated under strictly chronological temporal shift.

Minimal paper narrative

Lead with the empirical strength of full online scaling, introduce OPS once as prior work, and focus the method section on adaptive persistence. The no-slope and single-memory ablations support exactly two mechanism claims: slope flexibility matters, and adaptive memory matters. Discarded optimizer or memory variants need not appear as separate paper methods.

References

- C. Gupta and A. Ramdas. "Online Platt Scaling with Calibrating." ICML, 2023.
- E. Hazan, A. Agarwal, and S. Kale. "Logarithmic Regret Algorithms for Online Convex Optimization." Machine Learning, 69:169-192, 2007.
- M. Herbster and M. K. Warmuth. "Tracking the Best Expert." Machine Learning, 32:151-178, 1998.